

Cost-Sensitive Triage of Supplier Bank-Change Requests: A One-Step Bayesian Agent on 40 Simulated Cases*

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Abstract

A supplier bank-change request arrives with its hidden state unknown: the supplier’s genuine intent or fraud from a compromised channel. We build a POMDP-inspired one-step agent that maintains a Bayesian belief over {Genuine, Fraud} and selects among APPROVE, VERIFY (out-of-band callback), and ESCALATE (human controller) by expected cost. Evaluated blind on 40 simulated labeled cases (32 genuine, 8 fraud) against a rule-checklist agent, an email-authentication baseline, and two reference ceilings, the Bayesian policy catches all 8 frauds at \$4,200 total cost versus \$143,374 for the baseline. Accuracy alone (0.85 vs. 0.80) hides a \$139k cost gap. Likelihoods and costs are stated assumptions with a sensitivity procedure, not measured facts.

1 Introduction

The agent reads a supplier bank-change request. It must approve, verify via a second channel, or escalate to a human, because it cannot yet tell whether the request reflects the supplier’s genuine intent. Email authentication passes even on a fully compromised mailbox, so it cannot carry the decision [r/sysadmin contributors, 2026]. A wrong approval loses the wire irreversibly; a wrong check costs only callback labor and delay. The contribution is a tested, cost-sensitive triage policy: full predictions, multi-metric scoring beyond accuracy, six named failure modes, and repeat instructions, all on one fixed 40-case simulation.

2 Related Work and User Discussions

Practitioner discussions shaped every design choice. Operators report that verify-everything policies collapse under volume into rubber-stamping, and that calm, long-game fraud with no urgency signal is where losses concentrate [r/AL_Agents contributors, 2026]. Bank-verification APIs are judged a supplement that does not replace callback verification for bank-change events [r/fintechdev contributors, 2026].

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Practitioners verify changes by voice on a pre-known number, confirm by email to a known address, and inherit \$5–10k counter-signature materiality thresholds [r/Accounting contributors, 2026]. Authentication passes clean on mailbox compromise, leaving behavioral signals and procedural voice verification as the closing control [r/sysadmin contributors, 2026]. These findings set the evidence set, the callback-first action, and the weak weight on email authentication.

3 Agent Design

Input: one request record (callback status, first-time flag, urgency, reply-to drift, lookalike domain, tone deviation, email-auth status, amount). Hidden state: $S \in \{\text{Genuine, Fraud}\}$. Belief: $P(S \mid \text{evidence})$. Actions: APPROVE (pay), VERIFY (out-of-band callback, \$150 assumed), ESCALATE (controller hold and review, \$500 assumed). Cost: approving fraud loses the full amount; approving genuine costs \$0; checks cost their flat fees. Policy: the expected-cost rule of Section 5. Feedback: the callback outcome or controller verdict becomes new evidence for a belief update. The built-in human function is sending high-cost uncertain decisions to a human: posteriors at or above 0.65 escalate directly. This is the smallest testable version: a Naive Bayes update plus threshold code, no sequential planner.

4 Probability Model and Decision Rule

With prior $P(F) = 0.10$ (assumed base rate, not fitted to test labels) and a priori likelihoods $P(e_i \mid S)$ set from discussion direction, the posterior is

$$P(F \mid e_{1:k}) = \frac{P(F) \prod_i P(e_i \mid F)}{P(F) \prod_i P(e_i \mid F) + P(G) \prod_i P(e_i \mid G)}, \quad (1)$$

with $P(G \mid e_{1:k}) = 1 - P(F \mid e_{1:k})$, so beliefs always sum to 100%. Expected costs at posterior p and amount A are

$$\begin{aligned} E[C \mid \text{APPROVE}] &= p \cdot A, \\ E[C \mid \text{VERIFY}] &= 150, \\ E[C \mid \text{ESCALATE}] &= 500. \end{aligned} \quad (2)$$

Decision rule: ESCALATE if $p \geq 0.65$; APPROVE if $p \cdot A < 150$ and $p < 0.35$; otherwise VERIFY; high-value guard escalates when $A > 20,000$ and $p \geq 0.25$. Example (case C030, checked): likelihood products give $p = 0.3613$ via

Table 1: Blinded results on 40 simulated cases.

Policy	TP/FP/TN/FN	Prec	Rec	Rev.	Cost \$
Bayesian	8/6/26/0	0.571	1.000	0.350	4,200
Checklist	8/13/19/0	0.381	1.000	0.525	6,300
SPF baseline	1/1/31/7	0.500	0.125	0.050	143,374
Always approve	0/0/32/8	0.000	0.000	0.000	150,736
Always verify	8/32/0/0	0.200	1.000	1.000	6,000

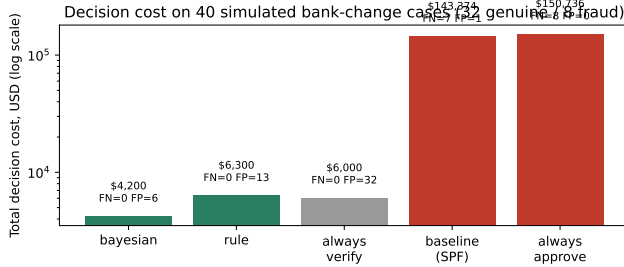


Figure 1: Total decision cost per policy (log scale), annotated with false negatives (FN) and false positives (FP). The Bayesian policy pays bounded check costs to avoid all wire losses.

Eq. 1; Eq. 2 gives \$2,768 vs. \$150 vs. \$500, hence VERIFY; a later failed callback updates the belief to 0.9314 and the action to ESCALATE. The true label is stripped before every policy call and used only for scoring.

5 Test Method

One fixed simulation of 40 labeled cases (32 genuine, 8 fraud) stored in `data/supplier_bank_change_simulated_dataset.xlsx` [Student Project, 2026]. Two agent policies (Bayesian cost-sensitive; deterministic red-flag checklist with zero weight on email authentication) plus an email-auth baseline (approve iff authentication passes) and two reference ceilings (always-approve, always-verify). Metrics beyond accuracy: confusion matrix, precision, recall, false-positive and false-negative counts, human-review rate, decision cost, and binned calibration (ECE). Every case×policy prediction is saved in `results/predictions.csv` (200 rows). The manuscript is formatted with the official IJCAI-ECAI-2026 author kit [IJCAI Organization, 2026].

6 Results

Table 1 and Figure 1 summarize the blinded run.

Both agents reach recall 1.000 with zero fraud loss; the baseline misses 7 of 8 frauds (recall 0.125) because authentication passes on compromise. Calibration ECE is 0.038 for the Bayesian policy (bins hold only 1–5 cases, so indicative only). Every number in this section was re-verified against `results/metrics.json`.

7 Failure Analysis

Six examined errors, each named: Stolen-Callback Confirmation (fraud with a “confirmed” callback, \$54,729 at stake), Urgency-Free Sleeper (calm \$56,875 fraud), Repeat-Vendor

Camouflage (fraud on a repeat vendor), Stale-Contact False Alarm (genuine case with a failed callback, \$150 wasted), Legitimate-Urgency Over-Triage (genuine first-time urgent change escalated), Benign Anomaly Coincidence (genuine reply-to/tone drift). Highest cost: false APPROVE of fraud (false negative), at \$3k–\$57k per miss versus \$150–\$500 per false alarm, because wires are unrecoverable while delays are resolved by one callback — a 20–380× asymmetry that justifies tuning for recall 1.000.

8 Limitations, Ethics, and Human Control

Simulated single 40-case split with no holdout; invented likelihoods under a false independence assumption; hand-picked thresholds never swept; flat assumed costs with zero recovery; small-sample calibration; 35–52% review rates exceeding realistic human capacity (rubber-stamp risk); trusted-but-unverified master-data numbers; and a one-step rule rather than sequential planning. Ethics: all amounts are simulated and must never be quoted as real fraud data. Human control is structural, not advisory: uncertain high-cost cases escalate to a controller by policy, and number changes require a separate authorization path.

9 Conclusion and New Questions

Cost-sensitive triage with hidden labels, multi-metric scoring, and named failures turns a vague fraud problem into a testable policy choice. Open questions: sweeping thresholds on a validation split; recovery-rate and capacity-budgeted costs; joint likelihoods for correlated signals; and a live pilot with audit logging. Sensitivity procedure: perturb each likelihood ratio $\pm 50\%$ (halve the failed-callback ratio $30 \rightarrow 15$) and confirm the zero-false-negative policy survives.

AI Use Statement

Muse Spark (via OpenCode) drafted code, computed metrics, formatted LaTeX, and acted as practitioner, probability, and preprint reviewer; 17 of 21 review comments were accepted with reasons in `review-record.md`. All Reddit/X content is the student’s own questions and summaries; no AI conversation was fabricated; every test reported was executed by `experiments/run_experiment.py`.

Contribution

Solo student work. Problem selection, public discussions, agent design, software development, tests, citation checks, preprint sections, and PDF checks were all performed by the student with AI assistance as stated above.

References

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`dataset.xlsx`, 2026. Simulated data; proportions are
directional assumptions, not measured statistics.